Comprehensive Analysis of Auxiliary Electric Motors in European and Asian Battery Electric Vehicles (2024-2026 Models)

Report Date: 2026-02-17

Executive Summary

This report presents a comprehensive technical, quantitative, and supply chain analysis of non-drivetrain auxiliary electric motors in Battery Electric Vehicles (BEVs) projected for the European market for the 2024-2026 model years. The research focuses on three distinct motor categories, now updated to reflect their specific roles in modern vehicle architectures: small stepper motors with a power rating under 50 watts, medium DC motors rated between 50 and 500 watts, and medium stepper motors with a power rating of 50 to 200 watts. The increasing sophistication of comfort, safety, and thermal management systems in BEVs has driven a significant proliferation in the number and complexity of these motors. They perform critical functions ranging from passenger cabin climate control and power seat adjustment to the essential thermal management of the high-voltage battery and power electronics, and the precise actuation of advanced driver-assistance and aerodynamic systems.

This updated analysis properly distinguishes and allocates applications between these motor categories. Small stepper motors are primarily utilized for lower-force, high-precision tasks such as the majority of HVAC blend door actuators [105, 106]. Medium DC motors are the workhorses for continuous power applications, including coolant pumps, window lifts, and seat adjustment mechanisms [88, 93]. Crucially, this report now identifies and details the significant role of medium stepper motors, which are now understood to comprise approximately 20% of the medium-power auxiliary motor segment. These motors are predominantly used in applications requiring a combination of precision and higher force, such as active grille shutters, adaptive headlamp leveling and swiveling systems, and charging port door actuators [109, 113, 118].

The report further provides a detailed quantitative breakdown of motor counts across six vehicle segments, from A-segment mini cars to F-segment luxury vehicles, revealing a direct correlation between a vehicle's market position and its auxiliary motor content. A-segment vehicles may contain as few as 10 motors, while high-end F-segment models can feature over 80 individual auxiliary motors.

Finally, the analysis delves into the material composition, cost structures, and supply chain dynamics for each motor type. Key materials such as copper, electrical steel, aluminum, and permanent magnets are examined. The report highlights the critical influence of raw material costs, particularly the volatile prices of copper and Neodymium Iron Boron (NdFeB) rare earth magnets, on the overall cost of these components [74, 98]. A significant finding is the automotive industry's profound supply chain vulnerability stemming from its dependence on China for the mining, processing, and manufacturing of rare earth magnets [81]. In response, global automakers and governments are actively pursuing risk mitigation strategies, including supply chain diversification, vertical integration, and research into alternative motor technologies, to secure the future of electric mobility [81, 84].

Introduction

The automotive industry's paradigm shift towards battery electric vehicles (BEVs) encompasses a transformation that extends far beyond the primary propulsion system. A fundamental, yet frequently underestimated, aspect of this evolution is the comprehensive electrification of auxiliary vehicle systems. In a BEV, every function that once drew mechanical, hydraulic, or vacuum power from a running internal combustion engine must now be powered by a dedicated electric motor. This necessity has led to the integration of a vast and diverse array of small and medium-sized electric motors, which are now responsible for a spectrum of functions essential to vehicle safety, passenger comfort, operational efficiency, and brand differentiation [88, 93]. The energy consumed by these numerous auxiliary systems has a direct and measurable impact on the vehicle's overall driving range, elevating the efficiency, weight, and control of their motors to a position of critical importance for automotive engineers and designers.

This report delivers an in-depth investigation into these auxiliary electric motors, specifically within the context of European and Asian BEV models anticipated for sale in the European market for the 2024-2026 model years. The analysis has been updated and structured around three precisely defined motor classifications to reflect the latest understanding of their application within the vehicle architecture: small stepper motors (<50W), medium DC motors (50-500W), and medium stepper motors (50-200W). For each of these distinct categories, this report provides a detailed examination of their specific applications, typical in-vehicle locations, core technical characteristics, and representative power ratings. By synthesizing and expanding upon information from technical reviews, component specifications, industry publications, and market analysis, this document provides an authoritative and holistic overview for stakeholders seeking to understand the component-level architecture, quantitative distribution, and material supply chain of modern electric vehicles. This updated framework, which now correctly identifies the significant role of medium stepper motors, offers a more accurate and nuanced perspective on the intricate electromechanical ecosystem that defines the contemporary BEV.

Part 1: Technical Analysis of Auxiliary Motor Applications

The functionality of a modern BEV is supported by a distributed network of electric motors, each selected for its unique performance characteristics. This section provides a detailed technical analysis of the three primary categories of auxiliary motors, outlining their core principles, key applications, and the engineering rationale for their deployment in specific vehicle systems. The updated analysis correctly attributes applications to each motor type, providing a clearer picture of the division of labor within the vehicle's electromechanical architecture.

Small Stepper Motors (<50 Watts)

Small stepper motors, characterized by power ratings typically below 50 watts, are foundational components for a multitude of precision control systems in modern BEVs. Their defining operational characteristic is the ability to rotate in discrete, fixed angular increments, known as "steps" [2, 4]. This allows for exceptionally accurate positioning of mechanical components without the need for complex and costly closed-loop feedback systems involving sensors like encoders or resolvers [3]. This principle, known as **open-loop control**, makes them remarkably reliable and cost-effective for applications where exact positioning and repeatability are more critical than high rotational speed or substantial torque [3]. In the energy-conscious design of a BEV, where every watt of consumption can affect the overall driving range, the inherent efficiency and low standby power draw of these motors represent a significant engineering advantage. They are most commonly employed in systems that require the precise and repeatable positioning of components such as flaps, valves, and indicator needles [6]. Their control is typically managed by dedicated driver circuits or integrated into larger electronic control units (ECUs) that communicate over vehicle data networks like the Local Interconnect Network (LIN) bus, enabling sophisticated, coordinated, and diagnostic-capable operation across the vehicle.

One of the most prevalent and high-volume applications for small stepper motors is within the Heating, Ventilation, and Air Conditioning (HVAC) system [103, 105]. These motors function as the primary actuators, driving the various doors and flaps that control air temperature, distribution, and intake source. Located deep within the main HVAC module, which is typically situated behind the vehicle's dashboard and firewall, these actuators are critical for creating and maintaining a comfortable cabin environment for passengers. A key function is the operation of the **blend door**, which regulates the final cabin temperature by precisely mixing hot air from the heater core with chilled air from the evaporator [105, 106]. A small stepper motor allows the vehicle's climate control unit to position this door with exceptional accuracy, enabling the system to achieve and maintain the exact temperature selected by the occupants. According to a 2019 review published by Transstellar Journals, the use of stepper motors in this role can reduce the energy consumption of the actuator system by as much as 20.15% compared to traditional DC motors, a vital energy saving in a BEV [106]. Beyond temperature regulation, small stepper motors also control the **mode doors**, which direct the flow of conditioned air to different cabin vents, such as the windscreen defroster, dashboard face vents, or floor-level footwell vents [105]. They also operate the **recirculation flap**, which switches the system's air intake between fresh outside air and recirculated cabin air. The precise, repeatable positioning offered by stepper motors ensures consistent and silent performance, allowing for complex climate control strategies, such as those found in vehicles from manufacturers like Mercedes-Benz, where efficient and inaudible operation is a key element of the premium passenger experience [104]. While the majority of these actuators fall into the sub-50-watt category, it is conceivable that in highly advanced multi-zone climate systems found in luxury vehicles, a specific high-force actuator could employ a medium stepper motor, though the bulk of the work is performed by their smaller counterparts.

Beyond HVAC systems, small stepper motors are also utilized in a variety of other low-power actuator roles throughout the vehicle. In some luxury vehicles, they are used for the automatic deployment and retraction of interior components, such as display screens or air vents that present themselves upon vehicle startup. Another application is within the instrument cluster itself, where small stepper motors are often used to drive the physical needles of analog-style gauges for speedometers, power-output meters, and state-of-charge indicators [6]. Their ability to move to and hold a precise position makes them ideal for providing clear and stable readings to the driver. Furthermore, in high-end E- and F-segment vehicles, the concept of automatic or powered doors may utilize small stepper motors for the latching and unlatching mechanisms, working in concert with larger DC motors that provide the primary force for opening and closing the door. In these applications, the stepper motor's role is one of precision locking and release, ensuring a secure and smooth operation. The power required for these slow, deliberate movements is minimal, placing these motors firmly in the small, sub-50-watt category.

Medium DC Motors (50 - 500 Watts)

Medium-power Direct Current (DC) motors, and particularly their more advanced brushless (BLDC) variants, serve as the robust workhorses in BEV auxiliary systems that demand continuous operation, high rotational speed, and significantly more power than small actuators can deliver. This category, defined for this report as motors with a power rating between 50 and 500 watts, is essential for a wide range of fluid-pumping and mechanical actuation applications [88, 93]. Unlike stepper motors, which excel at precise positioning, DC motors are engineered for sustained rotational speed and torque, making them the ideal choice for driving pumps, fans, and heavy mechanical linkages. In the context of a BEV, the absence of a traditional combustion engine necessitates a fully electrified approach to managing the substantial heat generated by the high-voltage battery, power electronics, and the electric drive motor itself, as well as providing heat for the passenger cabin. Efficient and reliable thermal management is not merely a matter of comfort; it is fundamental to battery health, performance, safety, and longevity. Similarly, convenience features that require significant force, such as moving windows and seats, rely on the power of these motors.

The most significant and critical application for medium DC motors in BEVs is driving the **electric water pumps** that circulate liquid coolant throughout the vehicle's various thermal management loops. A modern BEV architecture may feature multiple, independent cooling circuits to optimize the temperature of different subsystems. A primary and highly critical loop is dedicated to maintaining the high-voltage battery pack within its optimal operating temperature range, typically between 20°C and 40°C. Another separate or integrated loop may be responsible for managing the temperature of the power inverter and the main electric drive motor. A third circuit, often integrated with a sophisticated heat pump system, is responsible for cabin heating and cooling. Each of these circuits requires at least one, and often more, dedicated electric water pumps to ensure the continuous and reliable flow of coolant. These pumps are typically located in the engine bay or front trunk ("frunk") area, positioned near the components they service, or integrated into larger, consolidated thermal management modules. The power consumption of these pumps is a critical design parameter and varies based on the specific requirements of the cooling loop, such as the required fluid flow rate and system pressure. Analysis of electric pump specifications indicates that power ratings commonly fall squarely within the 50-500 watt range. For instance, data from Explorist.life on 12V systems shows that a pump drawing 7.5 amps consumes 90 watts [124], while other sources cite an average power consumption of 150 watts for a general-purpose water pump [123]. The total energy usage is a function of the pump's run time, which is dynamically managed by the vehicle's central thermal management controller. The pump may run continuously while the vehicle is in operation or cycle on and off as needed to maintain target temperatures, especially during high-stress events like DC fast charging, which generates significant thermal load. The use of efficient brushless DC motors is standard in this application due to their long lifespan, high reliability, and precise speed control capabilities, which allow the system to adjust coolant flow dynamically based on real-time thermal loads, thereby optimizing energy consumption and maximizing vehicle range.

Beyond thermal management, medium DC motors power a host of essential comfort and convenience features. **Power window lifts**, one for each door, use a geared DC motor to provide the necessary torque to raise and lower the window glass. Similarly, the **windshield wiper system** relies on one or two powerful DC motors to drive the wiper arms across the glass with sufficient force to clear rain and debris. In higher-segment vehicles, features like **power-folding mirrors**, **power-adjustable steering columns**, and **power liftgates or trunks** all utilize medium DC motors [88]. The most extensive use of these motors is in **power seats**. A basic 6-way adjustable seat may use three motors (for fore/aft, height, and recline), while an advanced 16-way massaging and ventilated seat found in a luxury vehicle could contain a dozen or more medium DC motors to control a vast array of adjustments, lumbar supports, bolsters, and massage functions. The proliferation of these features in C-segment vehicles and above is a primary driver of the increasing count of medium DC motors per vehicle.

Medium Stepper Motors (50 - 200 Watts)

The category of medium stepper motors, with power ratings between 50 and 200 watts, represents a critical and increasingly prevalent segment within modern BEV auxiliary systems. This updated analysis corrects previous assumptions and recognizes that these motors are the preferred solution for applications that require the high precision of a stepper motor combined with a level of force or torque that exceeds the capabilities of their smaller, sub-50-watt counterparts. These motors are almost exclusively of the hybrid synchronous design, which provides an optimal balance of torque, resolution, and holding power [68, 69]. While DC motors are suited for continuous rotation and small steppers are ideal for low-force positioning, medium steppers fill the crucial gap where precise, controlled, and forceful movement is paramount. Their ability to execute exact movements and hold a position firmly against an external load without a complex feedback system makes them highly reliable and cost-effective for a range of advanced safety and efficiency systems.

A key application for medium stepper motors is in **adaptive and automatic headlamp systems**. Modern BEVs are frequently equipped with automatic headlamp leveling and Adaptive Front-lighting Systems (AFS), both of which rely on stepper motors for their precise and rapid adjustments [109, 112]. These motors are housed directly within the headlamp assemblies. European regulations mandate automatic headlamp leveling for high-intensity headlights to prevent dazzling oncoming drivers [108]. The system uses vehicle level sensors to detect changes in the vehicle's pitch, and a control unit commands the stepper motors to make minute vertical adjustments to the headlamp angle. In more advanced AFS, stepper motors also provide horizontal, or swivel, adjustments, pivoting the headlamps into a turn based on steering angle and vehicle speed [109, 112]. The force required to move the entire projector and lens assembly quickly and accurately, especially in larger, heavier headlamp units, necessitates a motor with more torque than a small stepper can provide, placing this application firmly in the medium stepper category. Each headlamp in an advanced system can contain two or more of these motors, making them a significant contributor to the overall motor count in premium vehicles [111].

Another critical efficiency-related application is in **active grille shutters**. These systems, increasingly common on BEVs to optimize aerodynamic performance, feature motorized louvers in the front grille. The actuators that control these louvers must be robust enough to operate against significant air pressure at high vehicle speeds and must be watertight to withstand their exposed location [113]. A medium stepper motor provides the ideal combination of precise positional control—allowing the shutters to be partially or fully opened to fine-tune cooling and drag—and the torque required to ensure reliable operation under all conditions. By precisely controlling airflow, these systems directly contribute to optimizing the vehicle's energy consumption and extending its driving range.

Furthermore, medium stepper motors are the component of choice for **charging port door actuators**. This motorized mechanism is responsible for locking and unlocking the flap that covers the vehicle's charging port [118, 119]. The system must be highly reliable and provide a secure locking force to protect the high-voltage connection point. The action requires more force than a simple latch, needing to push the door open and pull it securely closed, often against ice or debris. A medium stepper motor provides the necessary torque and positional accuracy to ensure the door is either fully open or fully and securely latched, integrating with the vehicle's central locking system [121, 122]. In premium vehicles with dual charging ports (e.g., AC and DC), two such actuators may be present. The combination of these verified applications demonstrates that medium stepper motors are not a niche category, but rather an essential component class for enabling many of the advanced safety, efficiency, and convenience features that define the modern BEV.

Part 2: Quantitative Analysis of Motor Distribution by Vehicle Segment

The number and type of auxiliary electric motors integrated into a Battery Electric Vehicle are not uniform across the market. Instead, the motor count serves as a direct proxy for a vehicle's market segment, luxury level, and feature content. As manufacturers seek to differentiate their offerings and provide enhanced comfort, convenience, and safety, the quantity of electrified functions—and thus, the number of motors—increases dramatically. This section provides a quantitative analysis of auxiliary motor distribution across six standard European vehicle segments, from the entry-level A-segment to the flagship F-segment. The data presented is based on the updated understanding of motor applications, correctly allocating functions to small stepper, medium DC, and medium stepper motors, and provides a clear picture of the proliferation of these components as one moves up the automotive value chain.

The following tables summarize the estimated distribution of auxiliary motors. Table 1 provides a detailed breakdown of motor types used in specific vehicle systems and their estimated counts across the different segments. Table 2 provides a summarized total count for each motor category per segment, offering a clear overview of the overall trend. These estimates are derived from documented features of representative vehicles and the typical motorization strategies employed for those features.

TABLE 1: Comprehensive Auxiliary Motor Distribution by System and Vehicle Segment (2024-2026)

System/ Applica-tion	Motor Type	A-seg-ment	B-seg-ment	C-seg-ment	D-seg-ment	E-seg-ment	F-seg-ment
HVAC System	Small Stepper	4-6	4-6	6-8	6-8	8-12	10-15
Head-lamp System	Medium Stepper	2	2	2-4	4	4-6	6-8
Active Grille Shutter	Medium Stepper	0-1	1-2	1-2	2	2-4	4-6
Char-ging Port Lock(s)	Medium Stepper	1	1	1	1-2	2	2
Auto-matic/ Powere d Doors	Small Stepper	0	0	0	0	0-4	4
Window Lifts	Medium DC	2	4	4	4	4	4
Wind-shield Wipers	Medium DC	1-2	1-2	1-2	2	2	2
Coolant Pumps	Medium DC	1	1	1-2	2	2-3	3-4
Power Folding Mirrors	Medium DC	0	2	2	2	2	2
Power Seats (per seat)	Medium DC	0	0	2-4	4-6	5-8	8-12
Total Power Seats	Medium DC	0	0	4-8	8-12	10-16	16-24
		0	0	1-2	2	2	2-4

System/ Application	Motor Type	A-segment	B-segment	C-segment	D-segment	E-segment	F-segment
Sunroof/ Panoramic Roof	Medium DC						
Power Liftgate/ Trunk	Medium DC	0	0	0	2	2	2
Soft-Close Doors	Medium DC	0	0	0	0-4	4	4
Power Steering Column	Medium DC	0	0	0	0	2	2
Deployable Door Handles	Medium DC	0	0	0	0	4	4

TABLE 2: Summary of Total Estimated Auxiliary Motor Counts per Vehicle Segment

Segment	Small Stepper Motors	Medium DC Motors	Medium Stepper Motors	Total Motors (Range)
A-segment	4-6	3-4	3-4	**10-14**
B-segment	4-6	7-8	4-5	**15-19**
C-segment	6-8	8-14	4-7	**18-29**
D-segment	6-8	14-20	7-8	**27-36**
E-segment	8-16	22-35	8-12	**38-63**
F-segment	10-19	31-46	12-16	**53-81**

Analysis of Motor Distribution Across Segments

The data clearly illustrates a steep and consistent increase in auxiliary motor content as vehicle size, price, and feature complexity grow. In the **A-Segment**, representing mini cars like the Fiat 500e, the focus is on cost-effective essentials. The total motor count ranges from 10 to 14 units. This includes a basic HVAC system with 4-6 small stepper motors for blend and mode doors, two medium DC motors for the front window lifts, and one or two for the wipers. The thermal management system is typically simpler, requiring only a single medium DC coolant pump. Critically, even these basic vehicles require medium stepper motors for mandatory functions like headlamp leveling (2 units) and the charging port lock (1 unit), establishing a baseline presence for this motor type across all segments.

Moving to the **B-Segment**, which includes popular small cars like the VW ID.3 and Peugeot e-208, the total motor count rises to a range of 15 to 19. The primary increase comes from the addition of medium DC motors. These models typically feature four power windows instead of two, and often include power-folding mirrors, adding four more medium DC motors compared to their A-segment counterparts. The count of medium stepper motors also begins to climb, with the more frequent inclusion of active grille shutters (1-2 units) for improved aerodynamic efficiency, a feature that becomes increasingly standard in this class. The small stepper motor count for the HVAC system remains largely the same, reflecting a standard single-zone climate control system.

The **C-Segment**, or compact car class, which includes high-volume models like the Tesla Model 3 and Nissan Ariya, marks a significant inflection point where optional luxury features become more common. The total motor count expands to a range of 18 to 29. Here, we see the introduction of optional power seats, which can add 4 to 8 medium DC motors to the vehicle. Dual-zone climate control becomes more prevalent, increasing the number of small stepper motors in the HVAC system to between 6 and 8. The headlamp systems also become more advanced, with some models offering matrix or adaptive lighting that can increase the medium stepper motor count to 4. The potential inclusion of a motorized sunroof adds another one or two medium DC motors, contributing to the wide range in the total estimate for this segment.

In the **D-Segment** of mid-size cars, represented by models like the BMW i4 and Polestar 2, high-end features become more standard. The total motor count climbs to between 27 and 36. The most significant jump is in medium DC motors, driven by the standardization of more advanced power seats. A typical 12-way adjustable seat can use 6 motors, leading to a total of 12 motors for the front seats alone. Power liftgates become a common feature, adding another two DC motors. The medium stepper motor count also solidifies, with advanced adaptive headlamps (4 units) and more complex active grille shutters (2 units) becoming standard equipment. Some models may feature dual charging ports, increasing the actuator count to two.

The **E-Segment** and **F-Segment** represent the executive and luxury classes, respectively, with models like the Mercedes EQS, BMW iX, and Porsche Taycan. Here, the proliferation of motors is most extreme, as nearly every possible convenience and performance feature is motorized. In the E-segment, the total count ranges from 38 to 63 motors, while the F-segment can reach 53 to 81 motors or more. The number of small steppers grows to accommodate complex 4- or 5-zone climate control systems (up to 15 motors) and optional automatic doors. The count of medium DC motors explodes due to features like 16- or 24-way executive rear seats with massage and ventilation, panoramic sunroofs with multiple shades, soft-close doors, power-deployable door handles, and more complex thermal management systems requiring 3 or 4 coolant pumps. The medium stepper motor count also reaches its peak, with the most advanced adaptive headlamp systems (up to 8 motors) and multiple active aerodynamic components beyond just the grille shutters (up to 6 motors). This exponential growth in motorization in the luxury segments underscores the role of auxiliary motors as enablers of the premium, technology-rich experience that defines these vehicles.

Part 3: Material Composition and Supply Chain Analysis

The performance, cost, and availability of auxiliary electric motors are fundamentally tied to the materials from which they are constructed and the global supply chains that provide them. This section delves into the specific material composition, weight, critical alloys, cost structures, and supply chain dynamics for each of the three motor categories. The analysis reveals a complex interplay between material science, manufacturing economics, and significant geopolitical risks, particularly concerning the sourcing of critical raw materials like copper and rare earth elements.

Small Stepper Motors (<50W)

Small stepper motors are marvels of electromechanical precision, and their construction reflects a focus on achieving accurate motion in a compact and cost-effective package. Their material composition is a carefully balanced selection of metals, magnets, and plastics. The stationary part of the motor, the stator, is built around a core of laminated electrical steel, which is then wound with high-purity copper wire [4]. These copper coils generate the precise magnetic fields that drive the motor’s step-by-step rotation. The rotating part, or rotor, defines the motor’s type. In a simple Permanent Magnet (PM) stepper, the rotor is a magnetized cylinder of high-retentive steel [1, 4]. In a Variable Reluctance (VR) stepper, it is a toothed rotor made of soft iron [1, 4, 5]. However, the most common type in demanding automotive applications is the Hybrid stepper, which features a complex rotor with a permanently magnetized core, often using powerful rare earth magnets, encased in toothed soft iron cups [1, 4, 5]. The motor’s housing and structural components often utilize engineering plastics and aluminum to reduce weight and provide environmental protection. The weight of these small motors is typically low, with an estimated range of 0.2 kg to 0.4 kg, depending on the design and torque requirements [7, 8, 10, 11].

The performance of these motors is critically dependent on specialized alloys and magnet grades. The stator core is made from **non-grain oriented electrical steel (NOES)**, such as grades M19, M27, or M36 silicon steel [22, 25]. These grades are isotropic, meaning their magnetic properties are uniform in all directions, which is essential for the rotating magnetic fields in a motor [23, 25]. Thinner laminations, typically 0.20 mm to 0.35 mm, are used to minimize energy losses from eddy currents at higher operating frequencies [21, 22]. For high-performance hybrid steppers, the magnet of choice is a sintered **Neodymium Iron Boron (NdFeB)** magnet [28, 30]. These rare earth magnets offer the highest magnetic energy density available, enabling high torque in a small package [31, 33]. The specific grade, such as N48SH or N45UH, is critical [29, 31]. The number indicates magnetic strength, while the letters denote the maximum operating temperature. Automotive applications, with their harsh thermal environments, demand high-temperature grades (SH, UH, EH series, rated for 150°C to 200°C) to prevent irreversible demagnetization [29].

The cost structure of small stepper motors is heavily influenced by raw material prices and manufacturing volume. The most significant cost drivers are the copper for the windings and, for hybrid motors, the NdFeB magnet. Historical analysis shows that the magnet alone can account for a very large percentage of the total material cost, making the motor’s price highly sensitive to the volatile rare earth market [74]. Copper prices, which are also subject to significant global market fluctuations, represent another major variable cost [98, 100]. Manufacturing complexity, especially the intricate assembly of a hybrid rotor, also adds to the cost. The supply chain for these motors is global, with major manufacturers like Nidec Corporation and Johnson Electric operating production facilities worldwide [89, 90, 92]. However, this supply chain has a critical vulnerability: its overwhelming dependence on China, which controls approximately 90% of the global production of finished NdFeB magnets [81]. This concentration of supply creates a significant risk of price shocks and disruptions due to geopolitical factors. In response, automakers and governments are actively pursuing strategies to de-risk this dependency, including diversifying suppliers, investing in non-Chinese mining and processing operations, and funding research into rare-earth-free motor technologies [81].

Medium DC Motors (50-500W)

Medium DC motors are built for power and durability, and their material composition reflects this purpose. These motors are dominated by metals, with housings frequently constructed from die-cast **aluminum** for its excellent balance of low weight and high thermal conductivity, allowing the casing to act as an effective heat sink [53, 55]. In more rugged applications, **steel** may be used for the housing [53]. The electrical windings are universally made from **100% copper wire** to efficiently handle the higher currents required for this power class [48, 50]. The internal construction varies between brushed and brushless (BLDC) designs. Traditional brushed motors often use cost-effective **ferrite permanent magnets** in the stator, with the copper windings on the rotor [60, 62]. In contrast, high-performance BLDC motors typically place stronger **NdFeB rare earth magnets** on the rotor to achieve higher torque density and efficiency, with the windings in the stator [63, 65]. The weight of these motors is substantial, reflecting their power output. A 200W DC motor, for example, can weigh between 2.3 kg and 3.6 kg [41, 42, 43, 44, 45]. Copper content is significant, estimated to be between 15% and 18% of the total motor weight, with the remainder composed of steel, aluminum, and magnetic materials [13, 15].

The specific alloys and magnet grades are chosen to balance performance, cost, and longevity. The motor shaft, which transmits the mechanical power, is made from high-strength steel, such as **SAE 1045** carbon steel or, for more demanding applications, **4140** chrome-molybdenum alloy steel [57, 58]. The choice of magnet material represents a key engineering trade-off. **Ferrite magnets**, made from iron oxide and strontium carbonate, are inexpensive and highly resistant to high temperatures and corrosion, making them suitable for cost-sensitive applications where maximum power density is not the primary goal [60, 64]. For high-performance applications, however, **Neodymium (NdFeB) magnets** are preferred. They offer unmatched magnetic strength, enabling smaller, lighter, and more efficient motors, which is a critical advantage in BEVs [61, 65]. However, they are more expensive and require high-coercivity grades (e.g., SH, UH series) to withstand the high operating temperatures found in automotive environments without demagnetizing [60, 61].

The cost of medium DC motors is driven by this material selection. A high-performance BLDC motor using NdFeB magnets is significantly more expensive than a traditional brushed motor using ferrites, due to both the magnet cost and the required electronic controller [74]. The price of copper is a major factor; with a high copper content by weight, these motors are highly exposed to fluctuations in the copper market, which has seen sustained price increases due to global electrification demand [97, 99]. The supply chain for these motors is dominated by major Tier 1 automotive suppliers like Robert Bosch GmbH, Denso Corporation, and Continental AG [88, 90, 92]. These global giants operate extensive manufacturing networks to serve automakers regionally. While the copper supply chain presents challenges of price volatility, the most acute risk for high-performance BLDC motors is the same as for steppers: the dependence on the Chinese-controlled rare earth magnet supply chain [81]. The larger mass of magnetic material required for these more powerful motors makes this dependency even more significant, reinforcing the strategic imperative for the industry to diversify its sourcing of these critical materials.

Medium Stepper Motors (50-200W)

Medium stepper motors, which bridge the gap between low-force precision and high-power continuous motion, are constructed with materials designed to handle higher torque and thermal loads. Their material composition is an scaled-up version of the small hybrid stepper. The stator consists of a larger stack of electrical steel laminations and more substantial **copper windings** to carry the increased current needed for the 50-200W power rating [68, 70]. The rotor is a more robust hybrid design, featuring a powerful **sintered NdFeB permanent magnet** core to generate the high torque characteristic of this class [68, 69]. The housing is typically a rigid metal structure, often made of aluminum or steel, to provide structural support and manage heat dissipation. For applications requiring even higher force, these motors are often paired with a planetary gearbox, which introduces additional materials like **powder metallurgy** or machined **alloy steel** for the gears [70, 71]. The weight of these motors is considerably greater than their smaller counterparts, with an estimated range of 0.7 kg to 1.5 kg or more, especially for geared versions.

The alloys and magnet grades used in medium stepper motors are selected to maximize performance under higher stress. The stator laminations are made from high-quality **non-grain oriented electrical steel (NOES)**, with grades like M27 or M36 and thin gauges (0.35mm or less) being preferred to minimize core losses and improve efficiency under higher loads [22, 25]. The permanent magnet is the most critical component for performance. To produce the required torque and resist demagnetization from strong magnetic fields and high operating temperatures, high-strength, high-coercivity **NdFeB magnet grades** are essential [28, 30]. Grades such as **48SH**, **45UH**, or **42EH** are likely choices, offering both a powerful magnetic field and superior thermal stability with operating temperatures from 150°C to 200°C [29]. This thermal resilience is non-negotiable for ensuring the motor's long-term performance and reliability in the demanding automotive environment.

The cost of medium stepper motors is substantially higher than that of small steppers, driven directly by the larger quantity and higher quality of the materials used. The primary cost drivers are the larger mass of copper, the larger stack of high-grade electrical steel, and most significantly, the larger and higher-grade NdFeB permanent magnet [74]. The cost of the magnet increases exponentially with both size and thermal grade, making it the dominant component of the material cost [29]. The inclusion of a high-precision planetary gearbox adds another significant layer of cost, as the manufacturing of machined alloy steel gears is an expensive process [71]. The supply chain for these specialized motors is managed by global firms with expertise in precision mechatronics, such as Nidec, Johnson Electric, and AMETEK [89, 92]. This supply chain is critically dependent on the availability of high-grade NdFeB magnets, amplifying the industry's exposure to the risks associated with the China-dominated rare earth market [81]. The procurement of these powerful magnets is a key strategic challenge, and any volatility in the rare earth market has a direct and significant impact on the production cost and stability of these essential BEV components.

Synthesis and Conclusion

This comprehensive and updated analysis of auxiliary electric motors in 2024-2026 model year BEVs confirms their foundational role in the modern vehicle and reveals a sophisticated, multi-tiered ecosystem of electromechanical components. The investigation confirms the distinct and critical roles of small stepper motors, medium DC motors, and, crucially, medium stepper motors, each selected based on a careful engineering balance of precision, power, efficiency, and cost.

Small stepper motors, with power ratings under 50 watts, remain the component of choice for low-force, high-precision applications, most notably in the vast majority of HVAC system actuators [105, 106]. Their ability to provide accurate, repeatable open-loop control makes them indispensable for fine-tuning the cabin environment efficiently and silently.

Medium DC motors, particularly brushless variants in the 50 to 500-watt range, are the undisputed workhorses for continuous-duty and high-force applications. Their primary role in driving electric water pumps for the vehicle's complex thermal management systems is fundamental to the performance, longevity, and safety of the entire BEV platform [123, 124]. Furthermore, their proliferation in powering a vast array of convenience features, from power seats to window lifts, is a key driver of the dramatic increase in motor counts in mid- to high-segment vehicles.

This report's updated framework correctly identifies the significant and growing role of medium stepper motors (50-200W). This category is not a niche segment but a critical enabler of many advanced features. Their unique ability to combine high precision with significant force makes them the ideal solution for systems like adaptive headlamps, active grille shutters, and charging port locks [109, 113, 118]. Their presence is now understood to be standard across all vehicle segments, with their numbers increasing in premium models that feature more advanced lighting and aerodynamic systems.

Across all motor types that prioritize performance and compact size, a clear trend emerges: a deep reliance on advanced materials, particularly high-grade non-grain oriented electrical steel and, most importantly, high-strength Neodymium Iron Boron (NdFeB) permanent magnets [22, 29, 30]. This reliance, however, exposes the industry to significant challenges. The costs of two key materials—**copper** and **rare earth elements**—are major drivers of volatility [74, 97]. The price of copper, essential for all motor windings, is subject to sustained upward pressure from global electrification [99, 100]. More critically, the cost and availability of NdFeB magnets, which can dominate the material cost of a high-performance motor, are subject to the extreme price volatility and geopolitical risks inherent in the rare earth market [74, 81].

This leads to the report's most critical conclusion regarding the supply chain: the automotive industry's profound and systemic dependence on China for the processing of rare earth elements and the production of permanent magnets constitutes a strategic vulnerability of the highest order [81]. This single-point-of-failure risk threatens the stability and cost structure of BEV production worldwide. In response, a necessary and urgent strategic shift is underway. Automakers, suppliers, and governments are now actively collaborating to de-risk this critical supply chain through a multi-pronged approach that includes the diversification of sourcing, direct investment in non-Chinese mining and processing infrastructure, and long-term research into alternative magnet materials and rare-earth-free motor designs [81, 82, 84]. The future design, cost, and uninterrupted production of the hundreds of millions of auxiliary motors required for the transition to electric mobility will be inextricably linked to the successful navigation of these complex material and supply chain challenges.

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